Who Touched First? Detecting, Ordering and Placing Boot–Ball and Boot–Player Contacts from a Shoe-Mounted Inertial Sensor

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Abstract

The connected match ball detects the instant it is struck from an inertial sensor suspended at its centre. A foul is decided by the other half of that event: what the boot hit, and in which order. We ask whether a boot-mounted inertial measurement unit (IMU) can supply that half. Because no public recording of instrumented fouls exists, we build a physics-based simulator of the boot's signal — a running gait with foot strikes, a short stiff ball impact with post-impact ringing, a longer softer oblique player contact with a push tail, and their near-simultaneous superposition — passed through a sensor model (bandwidth, noise, bias, full-scale clipping, sampling rate), with real ankle-IMU recordings of walking, running and playing football (PAMAP2) as an alternative background. Four results follow. (i) A gradient-boosting classifier on interpretable pulse features separates {none, ball, player, both} at macro-F1 0.969 on synthetic and 0.908 on real background, against 0.547–0.666 for the ball-style threshold rule; a ± 16 g phone-grade part clips 66% of ball impacts yet still reaches 0.961. (ii) The ball-contact onset is timed to 0.52 ms at 1 kHz (5.8 ms at 100 Hz), and combining it with the ball's own sensor orders a ball contact and a player contact correctly at $\geq 90\%$ once they are 20 ms apart. (iii) In the anomalous case where the boot hits ball and opponent inside one pulse width, a two-template non-negative least-squares separator recovers the order in 0.88 of cases at 10 ms and 1.00 at 20 ms, but is at chance below 3 ms — a physical limit, not an algorithmic one. (iv) Fusing boot posterior, ball-sensor timing, separator order and a knowledge base of situational base rates in a calibrated stacked model yields a *probability* that the ball was played first with accuracy 0.944 and Brier score 0.038 (prior alone: 0.667 / 0.218), from which a cost-weighted rule with a review band decides fouls at 3.4% error while referring 5% of cases. Finally, foot-mounted inertial navigation with zero-velocity updates and sparse ultra-wideband fixes places the contact within 0.12 m at 10 Hz, enough to make a penalty-area call wrong with probability 1.7% at 25 cm from the line. The pipeline is configuration-driven and reproduces in minutes on a laptop.

1 Introduction

Modern football already instruments one of the two bodies in every contact. The match ball used at the 2022 World Cup carries an inertial sensor suspended at its centre, sampled at 500 Hz, whose sudden change of acceleration marks the instant the ball is touched; that timestamp, fused with limb tracking from cameras, is what made semi-automated offside possible and what settled whether a striker's head had grazed a goal-bound cross [3]. The ball, in other words, knows *when* it was hit. It does not know *what* hit it, nor whether the boot that hit it had struck an opponent a few milliseconds earlier — which is precisely the question a referee must answer to call a foul.

This paper asks whether the other body in the contact can be instrumented in the same spirit. A small IMU in the boot (in the heel counter or under the insole) sees every impact the foot delivers. Three signatures matter: the *ball*, a short, hard, forward-axis deceleration (foot–ball contact lasts about ten milliseconds [9]) followed by a high-frequency ringing from the stiff boot–ball interface; the *player*, a longer, softer, oblique pulse as soft tissue compresses, followed by a low-frequency push as the bodies keep moving into each other; and the *ground*, a periodic, mostly vertical foot strike that is the boot's normal background. The interesting — and contested — case is the fourth: the boot hits ball and opponent almost at once, and the call turns on which came first.

We treat this as a measurement problem with a decision-theoretic ending, and we answer it by simulation. There is no public dataset of instrumented fouls, so we build a physics-based generator for the boot's signal and pass it through an explicit sensor model, using real ankle-IMU recordings of walking, running and playing football [10] as an alternative background to check that the synthetic gait is not doing the work. The study then runs in four stages:

1. **Detection.** Can the boot tell ball from player from ground, and at what sampling rate and sensor grade? We compare the threshold logic of a connected ball ("a sudden change of the acceleration vector") with

a gradient-boosting classifier on interpretable pulse features.

2. **Timing and ordering.** How precisely is a contact onset timed, and — combining the boot with the ball's own sensor — how far apart must a ball contact and a player contact be for their order to be recovered?
3. **The anomalous overlap.** When both pulses fall inside one pulse width, a two-template non-negative least-squares fit separates them; we map where it works and where the physics says nothing can.
4. **The probabilistic call.** A knowledge base of situational base rates (how often a sliding tackle plays the ball first) is fused with the boot posterior, the ball-sensor timing and the separator's order into a calibrated probability that the ball was played first, and a cost-weighted rule with a video-review band decides.

A fifth section places the contact in space: foot-mounted inertial navigation with zero-velocity updates [4] and sparse ultra-wideband anchor fixes, evaluated as the probability of putting a contact on the wrong side of the penalty-area line.

Our contribution is a quantified map of what a boot sensor can and cannot settle, not a product. The numbers are specific to a simulator whose assumptions we state and whose knobs we expose; every figure and every number in the text regenerates from one command.

2 Related work

Instrumented balls and officiating. The connected ball [3] embeds a 500 Hz IMU whose impact detection, fused with optical limb tracking, feeds semi-automated offside and has been used to adjudicate touches. Our ball-sensor model (a detected impact time with a clock-sync error and sample quantisation) is an abstraction of that device; the boot sensor is the missing complement.

Kicking biomechanics. Foot–ball contact in the instep kick lasts roughly 8–12 ms with foot speeds of 15–25 m/s [9, 8]; the foot decelerates by tens of g over that interval, which is why impact-rated accelerometers (±200 g) are used in kicking studies and why a ±16 g consumer part clips. These figures set the ball template in our simulator.

Inertial sensing and change detection. Sensor error models (white noise, bias, bandwidth) follow the standard inertial-navigation treatment [14]. Onset estimation uses a threshold crossing with sub-sample interpolation and a matched filter on the pulse's rising edge [12]; two-pulse separation is non-negative least squares [7] over a lag grid with a BIC-like penalty for the second pulse.

Activity recognition from wearable IMUs. PAMAP2 [10] provides ankle-mounted IMU recordings including running and playing football; we use it strictly as background under which synthetic contacts are injected. Gradient boosting [5] on hand-crafted features is the classical strong baseline for such windows.

Calibrated probabilities and decisions. Isotonic calibration [15], the Brier score [2] and expected calibration error are the standard tools for asking whether a probability is honest; stacked generalisation [13] is the standard fix when naively fused sources are correlated. The decision rule is Bayes-optimal thresholding under asymmetric costs with an abstention band, which is how a video-review protocol is naturally formalised.

Foot-mounted navigation. Zero-velocity updates during stance make shoe-mounted inertial navigation drift linearly rather than cubically [4, 11]; ultra-wideband anchors bound the remaining drift [1]. We combine both in a small Kalman filter [6] and score the error where it matters: at the contact, against a line on the pitch.

3 Method

3.1 Simulating the boot's signal

Signals are generated at 4 kHz in the boot's frame (x forward, y lateral, z sole normal) and then passed through the sensor model. The *gait* is a running stride (2.4–3.2 strides/s): smooth swing-phase acceleration, a vertical half-sine foot strike (6–14 g, 25–40 ms) with a braking component, foot pitch that rotates gravity in the sensor frame, and the corresponding angular rate. Three *events* are injected at a random instant:

- **ball**: a half-sine deceleration along $-x$ with peak 50–130 g and duration 7–12 ms [9], followed by a damped 150–260 Hz ringing at 18% of the peak;
- **player**: an oblique half-sine of 8–45 g lasting 18–55 ms, followed by a 0.5–3 g push tail over 60–160 ms, with a lateral twist on the gyroscope;
- **both**: a ball pulse and a player pulse separated by a signed gap (negative: the player was hit first).

Figure 1 shows one sampled window of each class.

3.2 Sensor model and real background

The sensor chain is a first-order low-pass at 45% of Nyquist, a constant bias, white noise scaled with $\sqrt{f_s}$, clipping at full scale, and decimation to the sampling rate f_s. Two parts are modelled: an impact-rated *sport* accelerometer (±200 g) and a phone-grade *consumer* one (±16 g). The *ball sensor* reports an impact time only

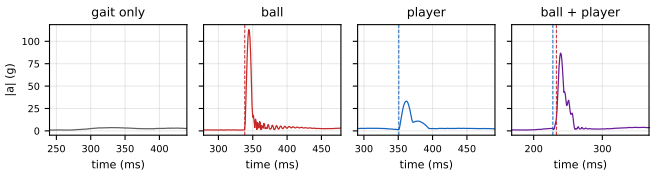


Figure 1: Acceleration magnitude of one window per class at 1 kHz (impact-rated sensor). Dashed lines mark the true ball (red) and player (blue) onsets. The ball is short, hard and rings; the player is longer, softer and pushes; in the overlap both sit inside one pulse width.

when the ball is struck, quantised to its 500 Hz grid with a 1 ms clock offset to the boot. As an alternative background we crop real 100 Hz ankle-IMU segments of walking, running and playing football from PAMAP2 [10] (10 subject–activity segments) and inject the same synthetic events; because those recordings contain nothing above 50 Hz, the real background is only used at its native rate.

3.3 Detectors

Features. Sixteen pulse descriptors of a window: peak $|\mathbf{a}|$ and peak jerk, widths at half and tenth maximum, high-frequency energy ratio and spectral centroid around the peak, zero crossings of the post-peak tail (ringing), direction cosines of the peak, gyroscope peaks, kurtosis, mean of the 40–120 ms tail (push), background RMS, and the number of secondary peaks.

Threshold rule. The connected-ball logic: contact if the peak exceeds τ; ball if the half-maximum width is below w, else player; both if a second comparable peak exists. (τ, w) are grid-searched on the training split, so this baseline is as strong as a threshold rule can be.

Boosting. Histogram gradient boosting [5] on the same features, returning class probabilities.

Onset. Two estimators anchored on the peak of the *horizontal* excess (foot strikes are vertical, contacts are not): the first sample above $\max(6\sigma, 0.08\,\text{peak})$ walking back from the peak, refined by linear interpolation; and a matched filter on the rising edge of a clean template with a parabolic sub-sample fit [12].

Separator. For the overlap, $|\mathbf{a}|$ − baseline is fit as $A\,T_{\text{ball}}(t - t_1) + B\,T_{\text{player}}(t - t_2)$, $A, B \geq 0$, over a lag grid, with the closed-form two-column non-negative least-squares solution [7] vectorised across all (t_1, t_2); a BIC-like penalty decides whether a second pulse is warranted. The output is the model (ball/player/both), both onsets, the signed gap and the order.

3.4 From evidence to a probability

For a challenge with situation s (standing tackle, sliding tackle, 50/50, late challenge) the *knowledge base* supplies a base rate $\pi_s = P(\text{ball first} \mid s)$; the rates used here are illustrative and are a configuration table. Three sensor sources produce a score in $[0, 1]$: the boosting posterior $P(\text{ball}) + \frac{1}{2}P(\text{both})$; the *timing* score $1 - \Phi\big((\hat{t}_{\text{ball}} - \hat{t}_{\text{boot,first}})/(\sigma\sqrt{2})\big)$ from the ball sensor's time against the boot's first onset, and 0 if the ball reports nothing; and the separator's order with confidence $\Phi(|\hat{\Delta}|/(\sigma\sqrt{2}))$. Each score is isotonically calibrated on a held-out half [15]. Two fusions are compared: the naive-Bayes log-odds sum $\operatorname{logit}\pi_s + \sum_k[\operatorname{logit} p_k - \operatorname{logit}\bar{p}]$, and a stacked logistic regression on the calibrated log-odds [13]. The *decision* calls "no foul" when $P(\text{ball first}) > c_{\text{miss}}/(c_{\text{false}} + c_{\text{miss}})$, and refers the case to review when the posterior lies within a band of that threshold.

3.5 Placing the contact

A 6-state Kalman filter on position and velocity integrates the world-frame acceleration obtained from gyro-integrated orientation, with zero-velocity updates whenever the foot looks planted ($|\boldsymbol{\omega}| < 40°/\text{s}$ and $|\mathbf{f}| \approx g$) [4, 11] and ultra-wideband position fixes ($\sigma = 0.10$ m) at a configurable rate. The trajectory is a 6 m/s run with a lateral cut over 6 s; error is scored over the second half of the run. The probability of misplacing a contact at distance d from a line is $\Phi(-d/\sigma_{\text{pos}})$.

4 Detection and timing

Classification. Figure 2 answers the first question. With 300 windows per class (840 train / 360 test), boosting reaches macro-F1 0.969 at 200 Hz and stays between 0.936 and 0.969 across 100–1000 Hz; the threshold rule, even with its two thresholds tuned, sits at 0.547–0.666. The gap is not about detecting *that* something was hit — both do that — but about *what*: a single peak-and-width rule cannot represent ringing, push tails, direction and secondary peaks at once. The phone-grade sensor clips 66% of the ball impacts, and at 100 Hz this costs it (F1 0.879), but by 500 Hz the shape of the clipped plateau, the ringing and the direction carry enough information to recover to 0.961: full scale matters less than sampling rate. On real ankle-IMU background at 100 Hz boosting reaches 0.908 (sport) and 0.874 (consumer), close to the synthetic values at the same rate, which says the synthetic gait is not making the task artificially easy; the threshold rule on real background is 0.630.

Timing. The ball onset is timed to 0.52 ms RMSE at 1 kHz, 2.15 ms at 500 Hz and 5.8 ms at 100 Hz; the matched filter's parabolic refinement beats the sample grid at high rates, the threshold crossing is better at low

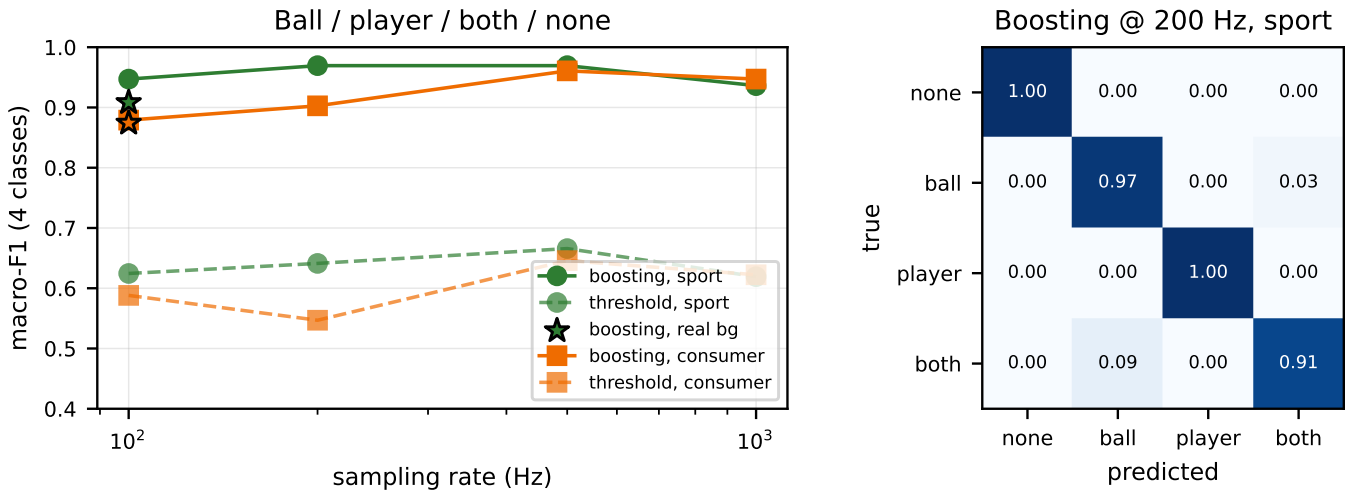


Figure 2: Left: four-class macro-F1 versus sampling rate for the boosting classifier (solid) and the tuned threshold rule (dashed), for an impact-rated (green) and a phone-grade ±16 g (orange) sensor on synthetic background; stars are boosting on real PAMAP2 ankle-IMU background at its native 100 Hz. Right: the boosting confusion matrix at its best rate. The only systematic confusion is *both* → *ball*: when the player pulse is faint the overlap looks like a clean kick.

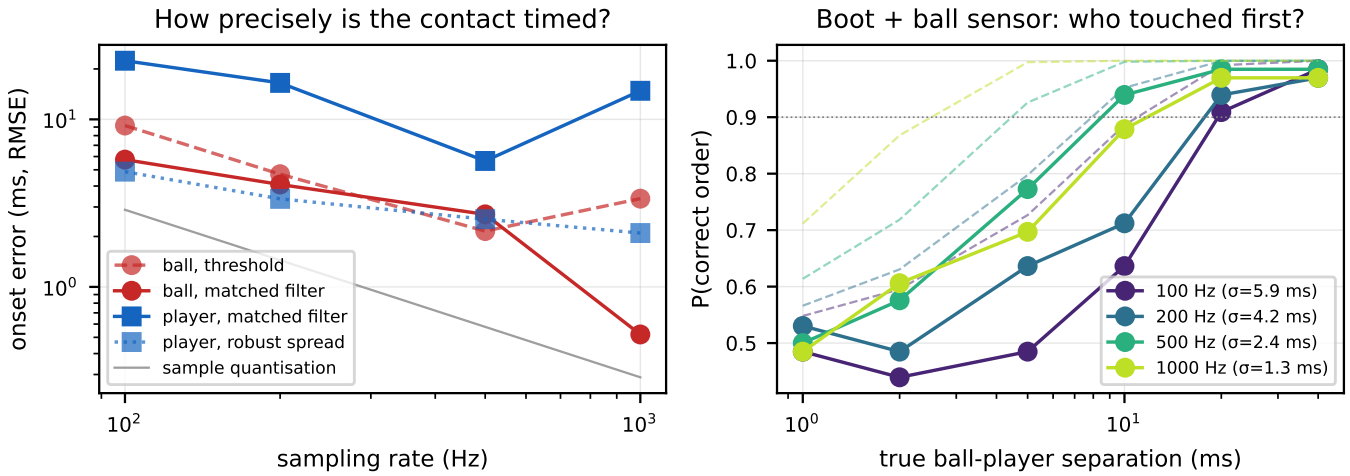


Figure 3: Left: onset error versus sampling rate (noise at nominal). The ball's onset follows sample quantisation down to sub-millisecond; the player's RMSE is dominated by a few gross errors (a faint contact mistaken for a foot strike), its robust spread is a few ms. Right: probability of correctly ordering a ball contact and a player contact from the boot's first onset and the ball sensor's time, versus their true separation, at each rate; dashed lines are the closed form $\Phi(\Delta/(\sigma\sqrt{2}))$ with the measured σ.

ones (Figure 3, left). The player onset is intrinsically softer: its robust spread is 2.1 ms at 1 kHz but its RMSE (14.8 ms) is dominated by a few percent of gross errors where a weak contact is out-shouted by a foot strike. Anchoring the estimator on the horizontal component of the acceleration (foot strikes are vertical) is what brings those gross errors from tens of percent to a few.

Who touched first, from two clocks. Given the boot's first onset and the ball sensor's time, "ball first" is declared when the ball fired within 2σ of the boot onset, where σ combines the boot timing error with the ball's clock offset and sample quantisation (1.2 ms). The right panel of Figure 3 plots the empirical probability of a correct order against the true separation. At 1 kHz the combined σ is 1.3 ms, the order is right in 0.88 of cases at 10 ms and passes 90% at 20 ms; at 100 Hz ($\sigma = 5.9$ ms) it is 0.64 at 10 ms. Below 2–3 ms every rate is at chance (0.61 at 2 ms, 1 kHz). The closed form with independent Gaussian errors is optimistic at small gaps — the two onsets are not independent when the pulses overlap, which is the subject of the next section — and matches well once the pulses are distinct.

5 The overlap, and the probabilistic call

The anomalous overlap. Figure 4 maps the case the project was started for: the boot hits the opponent and the ball inside one pulse width. With 120 windows per gap at 1000 Hz, the two-template separator declares two

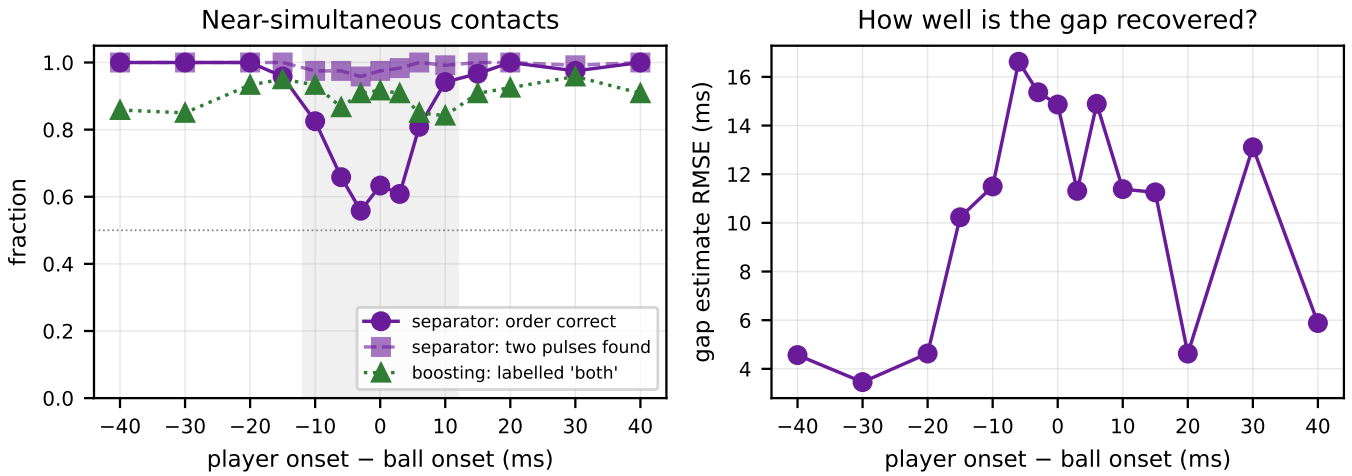


Figure 4: The anomalous case: ball and player struck within one window, versus the signed gap (player onset minus ball onset; negative = player first). Left: the separator finds two pulses almost always and orders them correctly outside the shaded band (one ball-pulse duration), while the boosting classifier labels the window *both* at a roughly constant rate regardless of gap. Right: RMSE of the recovered gap.

pulses in at least 0.96 of windows at every gap, and orders them correctly in 1.00 of cases at ±20 ms, 0.96 at ±15 ms, 0.88 at ±10 ms, 0.73 at ±6 ms and 0.58 at ±3 ms (0.63 at zero, which is chance by construction). The gap itself is recovered to 6 ms RMSE when the pulses are 20 ms or more apart and only to 15 ms inside the overlap. The boosting classifier, by contrast, labels 0.84–0.96 of these windows *both* at every gap: it knows two things were hit but has no notion of order. The shape of the left panel is the physical content of this paper: the ball pulse lasts 7–12 ms, so two contacts closer than that share their rising edges and no estimator can tell which began first from the boot alone. Above about one pulse duration the order is recoverable and above two it is essentially certain.

From evidence to a probability. Figure 5 reports the decision layer on 750 held-out challenges (half clean single contacts, half overlaps with gaps of 2–40 ms; base rate of ball-first 0.59). The knowledge-base prior alone is right 0.667 of the time with Brier 0.218: base rates are informative but not decisive. The boot classifier alone reaches 0.809 — essentially perfect on clean contacts (1.000) and near chance on overlaps (0.616), for the reason just given. Adding the ball sensor's timing is the step that matters: accuracy 0.951, Brier 0.049, with overlaps now at 0.930. Adding the separator's order under naive-Bayes fusion *hurts* slightly (0.927), because the boot posterior and the separator read the same window and are double-counted; the stacked logistic fusion resolves this (accuracy 0.944, Brier 0.038, ECE 0.031), learning weights of 1.68 (boot), 0.41 (ball timing), 1.29 (separator) and 0.83 (prior) on the calibrated log-odds. The reliability diagram shows the probabilities are honest across the range; the situation where the prior is most wrong (a late challenge, prior 0.78 accuracy) is where the sensors help most (0.96).

The call. With equal costs for a false foul and a missed foul the threshold is 0.5 and the error rate is 5.6%; adding a review band of ±0.15 refers 5% of challenges to video and lowers the error among those decided on the field to 3.4%. If a false foul is judged four times as costly as a missed one, the threshold moves to 0.20 and false fouls fall from 4.1% to 0.7% of challenges while missed fouls rise from 1.5% to 6.7%. The point of a calibrated probability is exactly that this trade is a policy setting, not a retraining.

6 Where was the contact?

Ordering the contacts says whether a foul happened; placing it says where the free kick is taken and whether it is a penalty. Figure 6 gives the boot's position error from its own IMU with zero-velocity updates (ZUPT) and sparse ultra-wideband (UWB) fixes, over 20 runs. Free inertial integration drifts to 3.3 m (sport IMU) and 7.6 m (consumer) horizontally within a few seconds; ZUPT alone brings this to 0.77 m and 1.98 m, because every stance resets velocity and drift accumulates per stride rather than per second [4]. One UWB fix per second is worth more than the IMU grade: 0.27 m for the sport part, and at 10 Hz 0.12 m, at 20 Hz 0.09 m (consumer: 0.12 m) — close to the 0.10 m of the anchors themselves, at which point the filter is interpolating between fixes with the IMU rather than dead-reckoning.

Read as a decision, the right panel is the penalty-area call. With 10 Hz fixes a contact 25 cm outside the line is placed inside with probability 1.7%, at 10 cm the probability is 20%, and at 50 cm it is below 0.01%; with ZUPT alone the 50 cm case is wrong 26% of the time.

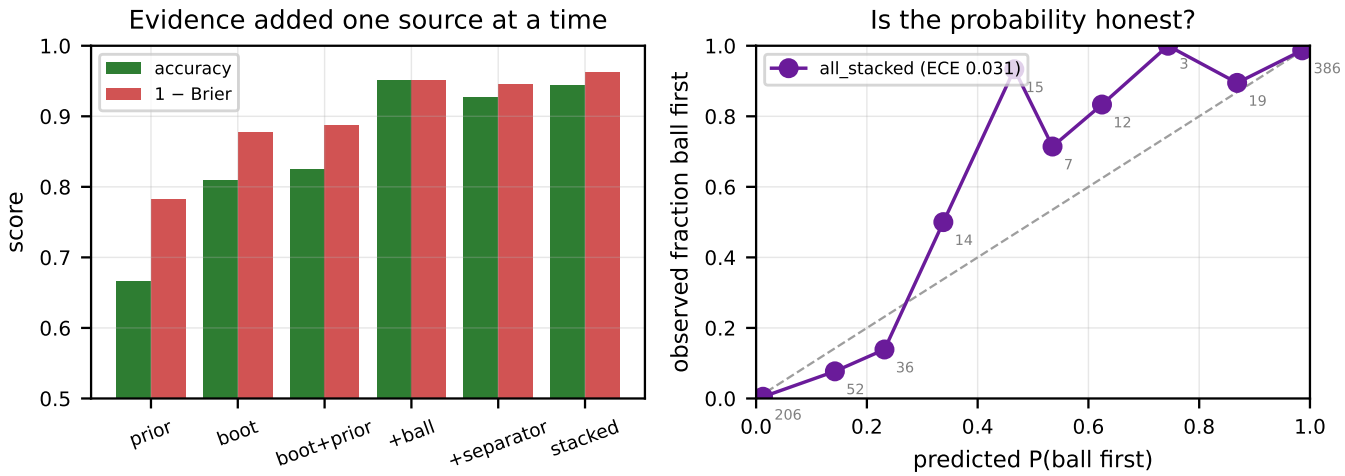


Figure 5: Left: accuracy and 1−Brier of P(ball first) as evidence is added — knowledge-base prior, boot classifier, ball-sensor timing, separator order — with naive-Bayes fusion, and the stacked logistic fusion of all sources. Right: reliability diagram of the best model (bin counts in grey).

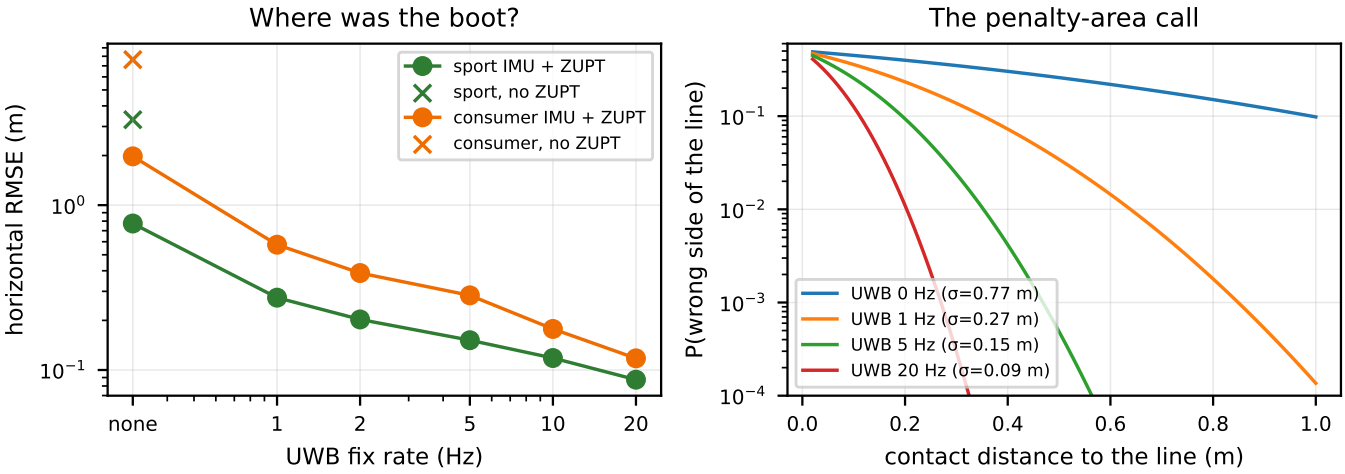


Figure 6: Left: horizontal position RMSE of the boot over the second half of a 6 s run with a cut, versus ultra-wideband fix rate, for an impact-rated and a phone-grade IMU with zero-velocity updates (crosses: free inertial integration). Right: probability of placing a contact on the wrong side of a line versus its distance to the line, for several fix rates.

This is the same logic as the timing result: there is a resolution (about σ_{pos} in space, about one pulse duration in time) inside which the sensors return a probability near one half, and outside which they return a confident answer. Note that the boot's position is a proxy: the foul's location is the contact point, which is within a boot length of the sensor, and the penalty-area rule concerns the *offence* location, so the same $\Phi(-d/\sigma)$ applies with d measured from that point.

7 Limitations and conclusion

Limitations. Everything here is simulated. The templates are half-sine pulses with parameters taken from kicking biomechanics, not measured boot signals from tackles; real player contacts come in more shapes (a shin, a stud on an ankle, a shoulder that never reaches the foot), some fouls involve no foot contact at all, and the boot sensor sees nothing of holding, pushing or simulation. The real PAMAP2 background guards against an easy synthetic gait but it is 100 Hz ankle data, not a boot at 1 kHz in a match. The knowledge base of situational base rates is illustrative; a deployed system would learn it from adjudicated matches, and it would also learn the templates and the separator's penalty. The ball-sensor model abstracts a real device to an onset time with a 1 ms clock offset; the actual figure sets the floor of the ordering curve directly, so it deserves measurement. The localisation study uses an idealised trajectory and a linear filter with no multipath or non-line-of-sight on the UWB. Every one of these is a configuration knob or a data swap, not a redesign, and the `gpu_full` configuration scales the sweeps and case counts by an order of magnitude.

Conclusion. A boot-mounted IMU can supply the half of the contact event that the connected ball cannot. The

boot reliably tells ball from player from ground (macro-F1 ≈ 0.969 synthetic, 0.908 on real background), and it does so with a phone-grade sensor once the sampling rate is high enough. It times the ball contact to 0.52 ms, which — with the ball's own clock — orders a ball touch and a player touch at 90% once they are 20 ms apart. In the contested overlap the two-template separator recovers the order down to about one ball-pulse duration (0.88 at 10 ms) and is at chance below 3 ms; that boundary is physics, and a referee's technology should report it as such — as a probability near one half — rather than as a verdict. The calibrated stacked fusion does exactly that, reaching 0.944 accuracy and Brier 0.038 against a base-rate prior at 0.667, and it turns the cost of a false foul versus a missed foul into a threshold and a review band rather than a retraining. Finally, ZUPT-aided inertial navigation with a few UWB fixes per second places the contact to about a tenth of a metre, enough for the penalty-area line except within a boot's width of it. The instrumented ball told us *when*; the instrumented boot can tell us *what*, *in which order*, *how sure*, and *where*.

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